UNDERSTANDING AND MAINTENANCE OF NETWORKS

Day 4: Network Cable Types and Tools

Q1. List 3 key differences between Ethernet cables and Fiber Optic cables. Give one real-world example of each.

Answer: Ethernet and Fiber Optic cables are both used to connect devices in a network, but they send data in different ways.

Ethernet Cable	Fiber Optic Cable
1. It sends data using electrical signals.	1. It sends data using light signals.
2. It is usually cheaper and easier to install.	2. Installation equipment and cable can cost more.
3. Copper Ethernet is commonly used for shorter links (typically up to 100 metres).	3. Fiber is suitable for much longer distances and high-speed links.

Real-world examples:

- Ethernet: Connecting a desktop computer or gaming PC to a home router or network switch.
- Fiber Optic: Connecting buildings across a college campus or linking equipment in a data center.

Q2. Draw and label the internal wiring of a straight-through Ethernet cable using the T568B standard.

Answer: A straight-through Ethernet cable has the same wiring order at both ends. In the T568B standard, the eight wires are arranged as follows:

Pin Number	Wire Colour (T568B)	Pin Number
1	White/Orange	1
2	Orange	2
3	White/Green	3
4	Blue	4
5	White/Blue	5
6	Green	6
7	White/Brown	7
8	Brown	8

Straight-through cable diagram:

END A – T568B	END B – T568B
Pin 1: White/Orange	Pin 1: White/Orange
Pin 2: Orange	Pin 2: Orange
Pin 3: White/Green	Pin 3: White/Green
Pin 4: Blue	Pin 4: Blue
Pin 5: White/Blue	Pin 5: White/Blue
Pin 6: Green	Pin 6: Green
Pin 7: White/Brown	Pin 7: White/Brown
Pin 8: Brown	Pin 8: Brown

Both ends use exactly the same colour order: 1 White/Orange, 2 Orange, 3 White/Green, 4 Blue, 5 White/Blue, 6 Green, 7 White/Brown, 8 Brown.

Q3. Write a step-by-step guide on how to attach an RJ45 connector to a CAT6 cable using an RJ45 crimper.

Answer: An RJ45 crimper is a hand tool used to press an RJ45 plug onto the end of an Ethernet cable.

1. Take a CAT6 cable, a CAT6-compatible RJ45 connector, a cable stripper and an RJ45 crimper.
2. Carefully remove about 2–3 cm of the cable's outer jacket without cutting the inner wires.
3. Separate the eight small wires and straighten them gently.
4. Arrange the wires in T568B order: White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown and Brown.
5. Keep the wires flat and in the correct order, then trim their ends evenly.
6. Slide all eight wires fully into the RJ45 connector. Check that each wire reaches the front of its channel and the cable jacket sits inside the plug.
7. Place the connector into the correct slot of the crimper and squeeze the handles firmly.
8. Repeat the same process at the other end if making a complete straight-through cable.
9. Use a LAN cable tester to check that all eight pins are connected in the correct order.

Result: The CAT6 cable is ready to be tested and used.

Q4. Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Which cable type and tool would you use? Explain in 3–4 sentences.

Answer: I would use CAT6 Ethernet cables to connect the 10 computers to a network switch. Ethernet gives a stable and fast connection, which is useful for LAN gaming, and it is affordable for a small café. I would use an RJ45 crimper with compatible RJ45 plugs to make the cable ends. Fiber optic cable and a fiber cleaver are not needed for these short connections between the PCs and the switch.